

Genomic Studies of Suicidal Thoughts and Behaviors in Latin American Populations: A Systematic Review

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ABSTRACT

BACKGROUND: Suicide is a leading cause of death globally, with disproportionately rising rates in Latin America. Despite increasing global interest in the biology of suicidal behavior, Latin American and Hispanic/Latinx populations remain underrepresented in genomic research. In this systematic review, we synthesized current evidence on genomic studies of suicidality in these populations.

METHODS: We reviewed molecular studies of suicidal ideation, suicide attempt, nonsuicidal self-injury, or suicide death among Latin American or Hispanic/Latinx individuals (PROSPERO registration: [CRD42023416616](https://doi.org/10.1016/j.biopsych.2025.12.009)). Eligible studies examined family risk, genetic, epigenetic, or transcriptomic markers. Searches were conducted in PubMed, SciELO, REDALyC, and regional journals. Synthesis used a narrative approach and random-effects meta-analyses when feasible. Risk of bias was assessed using the Newcastle-Ottawa Scale and the Q-Genie tool. A coauthorship network analysis was performed to characterize collaboration patterns.

RESULTS: Fifty-nine studies met inclusion criteria: 3 family, 50 genetic, 6 transcriptomic, and 3 epigenetic studies (categories not mutually exclusive). Candidate gene studies mainly examined serotonergic and stress-response pathways but yielded inconsistent results. Meta-analyses for *TPH1* (rs1800532), *HTR2A* rs6313 (T102C), and 5-HTTLPR (rs25531) showed no significant associations. Recent genome-wide and polygenic score studies identified ancestry-specific loci, although small samples and limited functional annotation resources hindered follow-up. Transcriptomic and epigenetic studies have implicated neurodevelopmental and immune-related processes. Collaboration networks were dominated by a single hub, with limited cross-group integration.

CONCLUSIONS: Genomic research on suicidality in Latin America is growing but remains fragmented and underpowered. Increasing representative sampling, implementing ancestry-aware methods, and fostering collaborations are essential to advance research and inform prevention strategies in these populations.

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Suicide is a major public health issue worldwide, causing approximately 700,000 deaths each year (1). In Latin America, suicide accounts for ~45,000 deaths annually (age-standardized rate of 5.23 of 100,000 inhabitants). Between 2010 and 2016, although global suicide rates declined by 9.8%, rates in Latin America increased by 6% (2,3). This increase was particularly pronounced between 1990 and 2019, with the largest increases in Paraguay (268.2%), Ecuador (245%), and Mexico (189%) (2). Consistent with global trends, suicide rates in Latin America are highest among older adults (>70 years) and markedly higher in men, with a male-to-female ratio of 3.28, significantly higher than the global ratio of 1.8 (4,5).

These increased rates have been linked to a complex interplay of cultural, behavioral, and socioeconomic factors, including poverty, violence, and inequality, highlighting the urgent need to better understand suicide in the Latin American context (4,6). Multiple factors contribute to suicidal behavior, including biological, psychological, and environmental influences (7,8). In recent years, there has been considerable progress in understanding the biological basis of suicidal behavior (7,9,10). However, most research in this area has focused on European ancestry populations, limiting its generalizability to other ancestry groups (11,12).

Environmental exposures and genetic architecture differ across populations. Latin American populations are highly

admixed, with diverse contributions from African, European, and Indigenous ancestries (13–15). This genomic diversity provides unique opportunities to uncover ancestry-specific mechanisms of suicidal behavior. However, Latin American and Hispanic/Latinx populations remain severely underrepresented in suicide genomics (16,17).

Here, we conducted a systematic review of studies investigating genomic variation in individuals with a history of self-injurious thoughts and behaviors, i.e., nonsuicidal self-injury (NSSI), suicidal ideation (SI), suicide attempt (SA), or suicide death (SD) from Latin American and Caribbean countries or individuals of Latin American ancestry worldwide. Although previous reviews have summarized genomic research on suicidal behavior globally (18), to our knowledge, none have focused on Latin American populations.

We also considered region-specific challenges, including the limited visibility of non-English scientific literature and the underuse of Latin American regional databases, which may have led to the omission of relevant studies in prior reviews (19). All eligible studies were included regardless of methodological limitations. In this systematic review, we aim to provide a comprehensive overview of current genomic research on suicidality in Latin American populations, identify knowledge gaps, and highlight opportunities for future research tailored to the region's distinct genetic and environmental context.

METHODS AND MATERIALS

Search Strategy

This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (20,21) and was registered in the PROSPERO database (ID: CRD42023416616) (22). We included studies with family risk, genetic, epigenetic, and transcriptomic analyses in individuals with SI, SA, SD, or NSSI from Latin American or Hispanic/Latinx populations. Searches were conducted from June 15 to July 15, 2024 in PubMed, SciELO, REDALyC, and regional Latin American psychiatry journals and were updated on October 15, 2025. We excluded duplicates, meta-analyses without new original data, case reports, narrative reviews, and conference abstracts. No language or year limits were applied. Full search terms, inclusion criteria, and journal lists are provided in [Supplemental Methods: Search Strategy and Data Sources](#).

Data Extraction and Risk of Bias Assessment

Screening and full-text review were performed independently by 2 reviewers, with disagreements resolved by a third reviewer. Data were extracted and cross-checked by independent reviewers using Rayyan (23). Risk of bias was assessed using the Newcastle-Ottawa Scale (NOS) (24) and the Q-Genie tool (25). For transcriptomic and DNA-methylation studies, a modified Q-Genie version was used (26). Detailed data extraction and bias assessment procedures are provided in [Supplemental Methods: Data Extraction and Risk of Bias](#).

Data Synthesis and Meta-Analysis

Extracted data were categorized by molecular phenotype (genetic, transcriptomic, or epigenetic), study type (e.g., genome-wide association study [GWAS], polygenic

score [PGS], gene expression, DNA methylation), and suicidality phenotype. Suicidality phenotypes were classified as SI, SA, SD, or NSSI, according to each study's definitions (see [Supplemental Methods: Phenotype Classification](#)). We conducted a narrative synthesis of key findings, summarized in tables for cross-study comparison. For candidate gene studies, we performed random effects meta-analysis when ≥ 2 studies examined the same variant using the *meta* package in R (version 4.4.3) (27). Quantitative analysis was performed for *TPH1* intron 7 (rs1800532), *HTR2A* rs6313 (T102C), and 5-HTTLPR (rs25531); remaining variants could not be meta-analyzed (see [Supplemental Methods: Meta-analysis of Candidate Gene Studies](#)) and were summarized narratively. A formal meta-analysis was not feasible for transcriptomic and DNA methylation datasets due to sample overlap and methodological heterogeneity (see [Supplemental Methods: Meta-analysis of Transcriptomic and Epigenetic Studies](#)).

Collaboration Network Analysis

A coauthorship network was generated to examine patterns of scientific collaboration in suicide genomics research in Latin America. Analytic procedures for network construction are described in [Supplemental Methods: Collaborative Network Analysis](#).

RESULTS

Search Results

Our initial broad search identified 745 records. After removing duplicates and screening titles, 57 studies met our inclusion criteria. The updated search identified 39 additional records, with 2 new studies included. The screening and selection process is summarized in [Figure S1](#), following PRISMA guidelines (20). Overall study quality was moderate (NOS mean = 6.34, SD = 1.81). Q-Genie scores indicated moderate methodological quality with considerable variability across studies (Q-genie mean = 56.75, SD = 10.8). A summary of the distribution of included studies by molecular phenotype, study type, and suicidality phenotype is shown in Table SA and illustrated in [Figure 1](#) and [Figure S2 \(Supplemental Results\)](#). Complete extracted information (including bias assessment) from included studies is available in [Tables S1](#) and [S3–S6](#).

Family Studies

One study showed strong associations between family history of SA and SA in probands (odds ratio [OR], 3.43; 95% CI, 1.88–6.27), including higher lethality (OR, 3.24; 95% CI, 1.53–6.86) and more violent methods (OR, 3.75; 95% CI, 1.68–8.45) (28). In another mixed psychiatric inpatient cohort, a family history of SA and SD also increased SA risk (OR, 2.34; 95% CI, 1.25–4.38) and was associated with more attempts over time, although lethality and method did not differ by family history (29). A third study of individuals with obsessive-compulsive disorder (OCD) reported an SA rate of 26% among probands ($n = 27$), but SA frequency did not differ between those with versus without a positive family history of OCD (30). A summary of family studies is provided in [Table 1 \(28–30\)](#), and full extracted study details are provided in [Table S1](#).

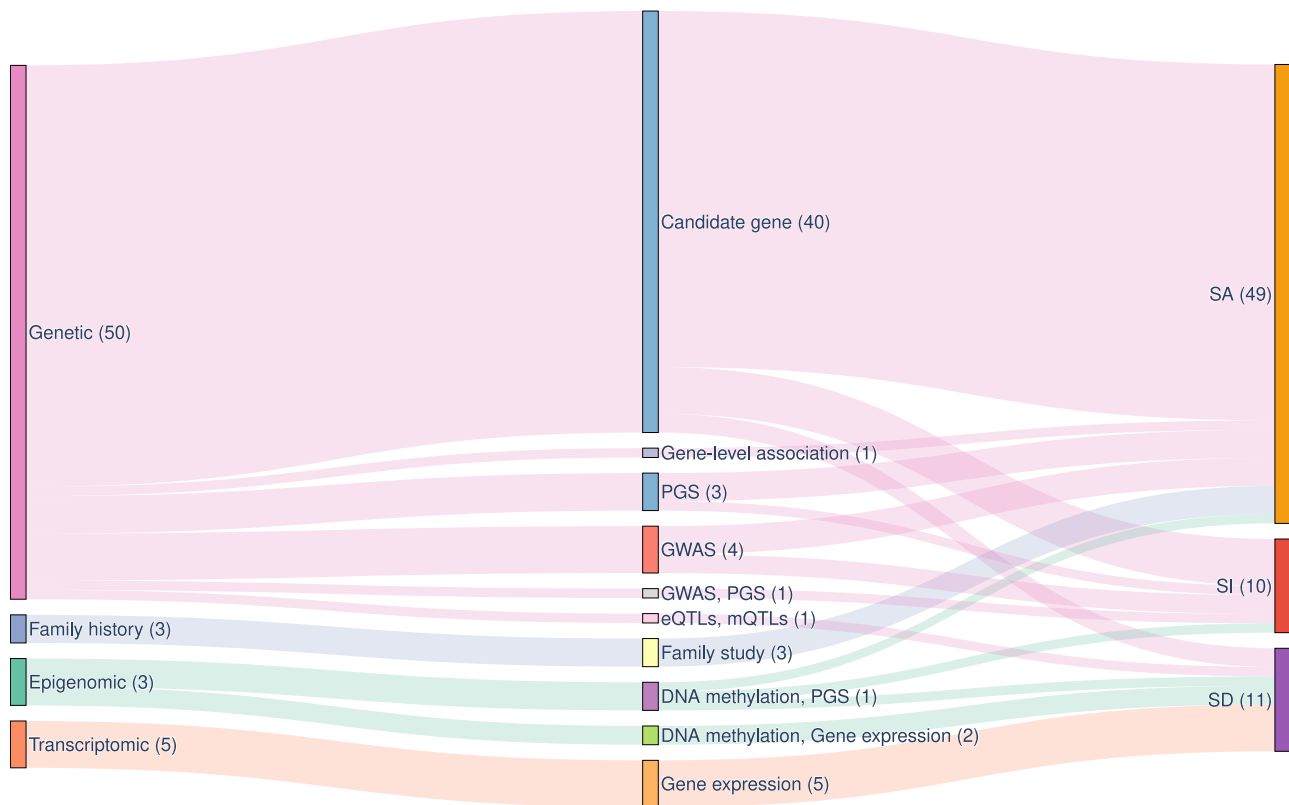


Figure 1. Sankey diagram showing study distribution across molecular level, study type, and suicidality phenotype. Each node represents a category, and link widths are proportional to the number of studies contributing to that category. On the left, studies are grouped by molecular level (genetic, transcriptomic, epigenomic, family history). The middle column displays study type (e.g., candidate gene, polygenic score, genome-wide association study). The right column indicates the suicidality phenotype evaluated: SA, SI, or SD. Studies could contribute to more than one phenotype category. Numbers in parentheses indicate the number of studies included in each category. eQTL, expression quantitative trait locus; GWAS, genome-wide association study; mQTL, methylation quantitative trait locus; PGS, polygenic score; SA, suicide attempt; SD, suicide death; SI, suicidal ideation.

Candidate Gene Studies

Candidate gene studies primarily focused on polymorphisms within neurotransmitter systems, particularly genes involved in the serotonergic system. Although some variants were associated with suicidal behaviors, findings were heterogeneous, likely due to methodological variability, small samples, and phenotype definition differences. Random effects meta-analyses did not identify significant effects for *TPH1* rs1800532 (OR, 1.39; 95% CI, 0.87–2.20; $p = .19$) or *HTR2A* rs6313 (OR, 1.06; 95% CI, 0.76–1.48; $p = .72$). Similarly, the primary 5-HTTLPR/rs25531 analysis comparing SA versus no SA did not show a significant association (OR, 0.49; 95% CI, 0.18–1.35; $p = .16$). In a secondary case-control analysis restricted to major depressive disorder (MDD) versus healthy control participants, 5-HTTLPR also showed no effect (OR, 0.94; 95% CI, 0.52–1.70; $p = .84$). Complete results from quantitative synthesis are shown in [Table S2](#). Detailed narrative information is shown in [Supplemental Results](#), with full extracted study details in [Table S3](#).

GWAS and PGS Studies

A gene-level study in Mexico (37 SA cases and 155 controls) identified an association between the *SCARA5* gene and SA (z -stat = 4.35; $p = 1.12 \times 10^{-6}$). Among 118 single nucleotide

polymorphisms (SNPs) showing significant associations, rs2685393 (located in *SCARA5*) showed the highest effect size (OR, 3.45; 95% CI, 1.82–6.57; $p = 1.58 \times 10^{-4}$), suggesting a role for *SCARA5* in SA in this population (31).

GWAS in the MVP (Million Veteran Program) and multi-ancestry meta-analyses identified genome-wide significant (GWS) loci in Hispanic/Latinx ancestry-stratified analyses. The MVP SA GWAS included 14,089 cases and 395,064 controls, of whom 7.4% of cases and 9% of controls were classified as Hispanic/Latinx. The following 3 GWS loci were detected in Hispanic-specific analyses: rs34173987 in *CACNA2D3* (OR, 2.34; 95% CI, 1.74–3.15; $p = 1.76 \times 10^{-8}$), rs6907713 near *TBC1D32* (OR, 15.67; 95% CI, 5.99–41.02; $p = 2.08 \times 10^{-8}$), and rs9596440 in *GUCY1B2* (OR, 2.73; 95% CI, 1.91–3.90; $p = 3.17 \times 10^{-8}$) (32). Follow-up PGS and genetic correlation analyses were restricted to European and African ancestry groups. Another U.S.-based analysis (121,211 cases/512,567 controls, including 13,230 Hispanic/Latinx cases) identified one GWS intergenic locus (chr4, rs116015815) associated with SI and SA in a Hispanic-specific GWAS (OR, 1.58; 95% CI, 1.35–1.85; $p = 3.47 \times 10^{-9}$) (33). Other ancestry-stratified analyses did not identify significant loci in Hispanic/Latinx (34,35) and Latin American populations (36), likely due to small sample sizes.

Table 1. Family-Risk Genetic Studies on Suicidal Behavior in Latin American Populations

Suicide Phenotype	Sample, Cases/Controls	Main Findings	Reference
SA	248 psychiatric patients/263 healthy control participants	Family history of SA was significantly associated with SA in the proband (OR, 3.43; 95% CI, 1.88–6.27), higher lethality (OR, 3.24; 95% CI, 1.53–6.86), and more violent methods (OR, 3.75; 95% CI, 1.68–8.45).	(28)
SA	237 psychiatric inpatients	Family history of suicidal behavior (SA and SD) increased the odds of SA (OR, 2.34; 95% CI, 1.25–4.38) and predicted more attempts over time; no differences were found in lethality/method.	(29)
SA	27 probands/286 first-degree relatives/187 second-degree relatives	Among probands with OCD, 26% had a history of SA. No difference was found in SA prevalence between those with vs. without a family history of OCD.	(30)

OCD, obsessive-compulsive disorder; OR, odds ratio; SA, suicide attempt; SD, suicide death.

The following 3 studies evaluated PGSs in Latin American or admixed populations: 1) a community-based study in Peru (510 cases/2788 controls) reported that an MDD PGS predicted SI (36); 2) in Mexico, a cohort of 1565 participants, including 1128 adolescents (ages 12–17 years) and 437 young adults (ages 18–25 years), showed that an MDD PGS was associated with SI and planning during adolescence but not in young adulthood (37); and 3) a multisite study across Mexico, the U.S., and Chile ($N = 2286$) reported that a history of SA was associated with greater pharmacotherapy exposure, although the genetic component of suicidality was not evaluated (38).

In the All of Us study, a bipolar disorder PGS was associated with SI (OR, 1.14; $p = .02$), and an MDD PGS was associated with SA (OR, 1.22; $p = .01$) among Hispanic/Latinx ancestry participants (39). Notably, PGSs were calculated separately for African/African American participants, admixed Hispanic/Latino American participants, and participants of European descent, accounting for population structure and enabling ancestry-specific linkage disequilibrium modeling.

Finally, an expression quantitative trait loci (eQTL) and methylation quantitative trait loci (mQTL) study conducted in Mexico ($N = 21$) identified 6224 mQTLs and 2239 eQTLs associated with SD (40). A summary of GWAS and PGS studies is provided in Table 2 (31–40), and full extracted study details are provided in Table S4.

Gene Expression Studies

Postmortem analyses with Latin American samples primarily sampled the prefrontal cortex (6 studies) and the hypothalamus (1 study). Two studies used candidate-gene approaches. One reported increased expression of *HTR2A* (2-fold; $p = .01$) in the dorsolateral prefrontal cortex and of *MAOA* in the hypothalamus (1.37-fold; $p = .021$) of individuals who died by suicide compared with control individuals (41). The other reported significantly higher *SKA2* expression in the dorsolateral prefrontal cortex of individuals who died by suicide ($p = .044$) and a significant association between elevated *SKA2* expression and the A allele of rs7208505 ($p = .027$).

The 3 remaining studies examined genome-wide transcriptional signatures in high-risk subgroups, such as individuals with substance use disorders (SUDs) or dual diagnoses (DDs), i.e., individuals with a SUD and a comorbid mood disorder. Among suicide decedents with a history of SUD, 550 differentially expressed genes (DEGs) were identified compared with nonsuicide decedents with SUD, primarily enriched in oxidative phosphorylation pathways (42). Another SUD-focused study identified 1794 DEGs in suicide decedents, with co-expression network analyses revealing hub genes with potential pharmacological relevance (43).

In individuals with DD who died by suicide, transcriptomic contrasts against those with SUD-only and MDD-only identified 255 DEGs (DD vs. SUD) and 172 DEGs (DD vs. MDD) (44), suggesting that comorbidity influences brain gene expression in Latin American samples. Another study aimed to better understand the biological basis of the gender paradox in suicide, wherein men die by suicide more frequently despite higher rates of SA among women (45). The study found sex-specific gene expression changes, with 1729 DEGs in females who died by suicide and 1997 DEGs in males who died by suicide compared with control participants (46). The study also reported a significant overlap between female- and male-specific DEGs, with 903 genes shared across both comparisons that were enriched for mitochondrial function, vesicular transport, ATPase (adenosine triphosphatase) activity, and cellular component assembly pathways, indicating partially convergent molecular pathways between sexes. A summary of gene expression studies is provided in Table SB, and full extracted study details are provided in Table S5.

Epigenetic Studies

Two postmortem brain studies integrated DNA methylation and gene expression in SD cases; one identified 22 CpG-expression correlations, largely related to neurodevelopment (17), and another detected 70 genes with concordant methylation-expression changes (parallel analyses of DNA methylation and gene expression) enriched for signal transduction, synaptic plasticity, stress hormone regulation, and immune response pathways (47).

A third integrative study combining PGS with DNA methylation data found 153 differentially methylated CpG sites between individuals with high and low genetic risk for SA, which were annotated to genes involved in neurodevelopment. These results suggest that genetic liability may shape epigenetic vulnerability to suicidality in this population (48). The top CpG site (cg19269713) was mapped to *ANXA4*

Table 2. GWAS and Polygenic Score Studies on Suicidal Behavior in Latin American Populations

Molecular Phenotype	Suicide Phenotype	Ethnicity or Ancestry of Participants	Sample Size, Cases/Controls	Main Results and Allele Frequency (If Reported)	Reference
Gene-Level Association Study	SA	All participants were born in Mexico and had Mexican ancestry for at least 2 generations.	37/155	SCARA5 gene was significantly associated with SA ($p = 6.77 \times 10^{-6}$); 118 SNPs were associated with SA.	(31)
GWAS	SI	Controls: African American (17.7%), Asian American (1.3%), European American (73.5%), Hispanic American (7.5%) Cases: African American (25.1%), Asian American (1.5%), European American (62.1%), Hispanic American (11.2%)	99,814/ 512,567	No significant association in ancestry-specific analysis in Hispanics. Meta-analysis identified 4 genome-wide significant loci for SI across ancestry groups in the MVP cohort.	(34)
GWAS	SA	Cases: African (18.8%), Asian (1.0%), European (72.7%), Hispanic (7.4%) Controls: African (24.9%), Asian (0.8%), European (65.3%), Hispanic (9.0%)	14,089/ 395,064	Three genome-wide significant loci were identified in the Hispanic ancestry-specific GWAS: rs34173987 in <i>CACNA2D3</i> (OR, 2.34; 95% CI, 1.74–3.15; $p = 1.76 \times 10^{-8}$), rs6907713 near <i>TBC1D32</i> (OR, 15.67; 95% CI, 5.99–41.02; $p = 2.08 \times 10^{-8}$), and rs9596440 in <i>GUCY1B2</i> (OR, 2.73; 95% CI, 1.91–3.90; $p = 3.17 \times 10^{-8}$).	(32)
GWAS	SA	Multiple (African, East Asian, European, Hispanic/Latino)	43,871/ 915,025	Hispanic/Latino participants were limited to 1 cohort and not meta-analyzed.	(35)
GWAS	SI, SA	Cases: African (25.0%), Asian (1.4%), European (62.7%), Hispanic (10.9%) Controls: African (17.7%), Asian (1.3%), European (73.5%), Hispanic (7.5%)	121,211/ 512,567	One genome-wide significant intergenic locus (chr4, rs116015815) associated with SI and SA in a Hispanic-specific GWAS (OR, 1.58; 95% CI, 1.35–1.85; $p = 3.47 \times 10^{-9}$). Effect allele (T) frequency: 0.01	(33)
GWAS, PGS	SI	Self-identified as Mestizo ethnicity (76.1%)	510/2788	MDD PGS significantly predicted suicidal ideation ($R^2 = 0.3\%$, $p = .01$).	(36)
PGS	SI, SA, suicide planning	Participants recruited in Mexico City; ancestry not reported	Adolescents: 1128 Young adults: 437	MDD PGS was associated with suicidal behavior during adolescence ($p = 7.1 \times 10^{-5}$), ideation ($p = 5.7 \times 10^{-5}$), and planning ($p = 3.3 \times 10^{-3}$).	(37)
PGS	SA	Hispanic (12.3%), non-Hispanic (87.7%)	2286	History of SA was associated with higher pharmacotherapy exposure ($p < 1 \times 10^{-4}$); PGSs were evaluated solely in relation to pharmacotherapy exposure, not suicidality outcomes.	(38)
PGS	SA	African American (4%), Hispanic or Latino (4.2%), more than 1 population (4.8%), White (83.1%)	41,379	BD PGS was associated with suicidal ideation (OR, 1.14; $p = .02$); MDD PGS was associated with SA (OR, 1.22; $p = .01$).	(39)
eQTLs and mQTLs	SD	Ancestry not assigned; principal component analysis indicated no evidence of population admixture	21	Identified 6224 mQTLs and 2239 eQTLs associated with SD in the prefrontal cortex.	(40)

BD, bipolar disorder; eQTL, expression quantitative trait locus; GWAS, genome-wide association study; MDD, major depressive disorder; mQTL, methylation quantitative trait locus; MVP, Million Veteran Program; OR, odds ratio; PGS, polygenic score; SA, suicide attempt; SD, suicide death; SI, suicidal ideation; SNP, single nucleotide polymorphism.

($\log_2$ fold change = -0.00684 ; $p = 2.25 \times 10^{-8}$). A summary of epigenetic studies is shown in Table SC, and full extracted study details are provided in Table S6.

Network Analysis

The bipartite author-publication network was highly centralized, dominated by a single giant component (Figure 2). This central cluster, densely interconnected and centered around a single publication node, consists largely of researchers based in the United States and Mexico who frequently coauthor articles together. Around this core were moderately sized clusters and numerous smaller, fragmented groups connected to only 1 or 2

publications. This configuration reflects a typical coauthorship pattern; many "one-shot" authors (individuals who participate in only one publication) are linked to a limited number of recurring investigators who act as bridges across publications and help maintain the overall cohesion of the system. Several peripheral groups functioned as isolated teams or single-institution collaborations, showing limited integration with the main collaborative hub and highlighting restricted cross-institutional and cross-national collaboration across the region.

In terms of publications, the degree distribution is shaped by a single extreme outlier (35), corresponding to a large-scale GWAS meta-analysis with more than 200 coauthors.

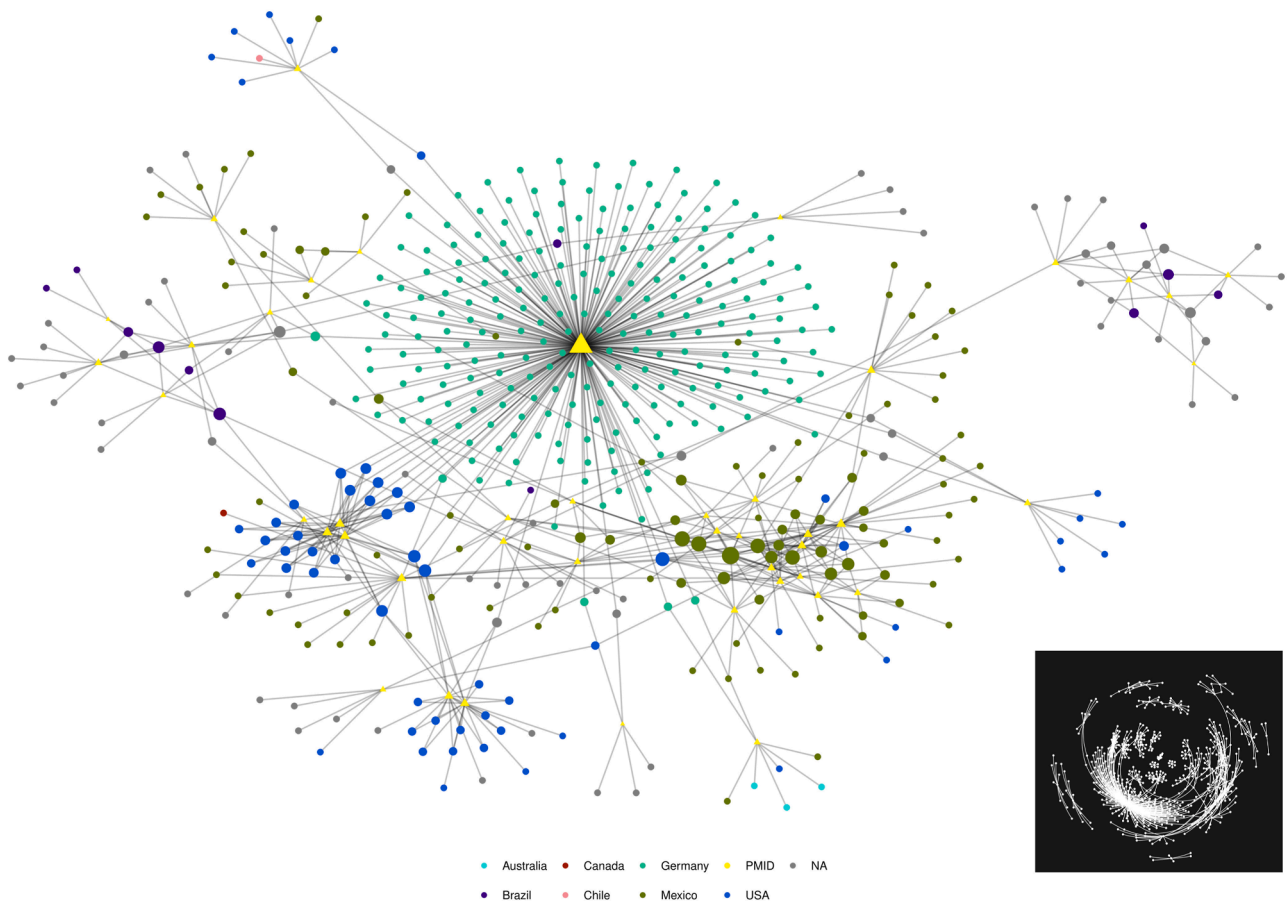


Figure 2. Bipartite collaboration network of authors and publications included in this systematic review. Each node represents either a publication (triangle) or an author (circle). Edges represent authorship links. Node color corresponds to the author's country of affiliation. The square inset in the lower right corner shows the same network without country colors, using the standardized circular layout provided as a visual reference. NA, not applicable; PMID, PubMed identifier.

Excluding this exceptional case, most articles included 7 to 22 authors, indicating moderate-scale collaborations that better represent the typical behavior of the network.

Centrality analysis identified a small group of authors as key international connectors that sustain the global structure. Walss-Bass (United States) shows the highest betweenness value and serves as the main vertex of the core; Krebs (Germany) represents the European axis associated with the massive meta-analysis; Nicolini (Mexico) acts as the bridge between the Latin American cluster and the central component; and Correa (Brazil) links the South American periphery to the main body of the network. Country affiliations should be interpreted cautiously because metadata often include multiple institutions per author. The interaction among these nodes defines the transnational architecture of the system, where international collaboration occurs in a mediated and asymmetrical fashion; connections between countries exist, but they are channeled through specific intermediaries rather than through a dense web of collaborations. Altogether, the observed structure suggests a hierarchical coauthorship system in which many peripheral participants depend on a few coordinating nuclei. This pattern limits horizontal knowledge

exchange and makes the network structurally dependent on a small number of bridging authors.

Complete results of the network analysis are available in [Table S7](#) (network metrics, degree and component size distributions) and [Figure S3](#) (plot of degree distributions).

DISCUSSION

This systematic review summarizes the current state of genomic research (genetic, epigenetic, transcriptomic) on suicidal behavior in Latin American and Hispanic/Latinx populations, including multi-ancestry studies that included individuals of Latin American descent. Despite the global expansion of research into the molecular mechanisms underlying suicidality, studies in Latin American populations remain scarce and underpowered compared with studies in European ancestry samples. This disparity is concerning given that suicide rates in Latin America are rising more rapidly than in many other parts of the world (4), highlighting the urgency of region-specific studies.

Three family-based studies were identified and collectively suggest that family history increases vulnerability for suicidal behavior in Latin American populations, including greater

lethality and more violent methods (28,29). Nonetheless, one study reported no difference by family history, likely reflecting limited power (30). The absence of twin/adoption designs in Latin American populations is noteworthy, as they limit separation of genetic versus shared environmental effects that cannot be captured by molecular studies alone (49).

Findings from candidate gene studies were generally inconsistent, although genes such as *BDNF*, *5-HTT (SLC6A4)*, *TPH1*, and *HTR2A* were repeatedly examined, reflecting a focus on serotonin- and stress-related pathways. This is consistent with early global research trends (18), particularly the longstanding interest in serotonergic mechanisms. Genetic associations were mainly inconsistent, likely due to small sample sizes, differences in phenotype assessment, methodological heterogeneity, limited replication, and inadequate adjustment for ancestry. In this meta-analysis, none of the tested variants showed statistically significant effects, consistent with recent critiques that candidate gene approaches are underpowered and rarely replicate (50). These issues have contributed to the field's shift toward genome-wide and polygenic approaches, which are better suited to capture the genetic architecture of psychiatric traits (51).

GWAS and PGS studies in Latin American samples have provided greater methodological rigor and begun to identify ancestry-specific risk loci, i.e., rs34173987, rs6907713, rs9596440, and rs116015815 (32,33). However, Latin American cohorts remain underrepresented in large-scale meta-analyses, limiting the generalizability of suicide genomic studies to this population. A recent MVP analysis examined PGSs for externalizing disorders across ancestries (52), but the Hispanic/Latinx subgroup analyses (7.5% of total participants) were underpowered, limiting detailed evaluation within this group. Although suicidality was not directly examined, this illustrates the potential of large-scale resources for future ancestry-specific analyses.

The lack of ancestral diversity in functional genomic datasets, such as tissue-specific gene expression and DNA methylation profiles, also constrains follow-up analyses (53) and limits statistical power to detect meaningful effects. These functional datasets are essential for the biological interpretation of genomic findings, yet most currently available resources are based predominantly on individuals of European ancestry (54). Developing functional genomic resources that reflect the genetic background of Latin American populations is critical to reducing bias and improving the accuracy, generalizability, and translational relevance of genomic research in these populations.

Many studies have examined U.S.-based Hispanic populations, which may not be generalizable to Latin America, where contributions from Indigenous African, American, and European ancestry vary widely, resulting in distinct genetic architectures (55,56). Environmental exposures also differ substantially, influenced by sociopolitical, economic, and cultural contexts that vary not only between countries but also within regions of the same country (56). This further emphasizes the importance of ancestry-aware methods and regionally representative sampling.

Encouragingly, recent studies in Peru, Mexico, and Chile applied PGS analyses to local cohorts, reporting associations between MDD PGSs and suicidal behaviors, particularly SI

and planning. However, PGS models based on GWASs conducted in European-descendant populations reduce their predictive accuracy in admixed and non-European populations (57), highlighting the need for ancestry-specific discovery datasets and more inclusive reference panels to improve polygenic risk prediction in diverse populations.

Transcriptomic studies revealed widespread differential brain gene expression between individuals who died by suicide and control individuals in Latin American samples. Subgroup analyses suggested that sex, psychiatric comorbidities, and SUD may particularly influence gene expression profiles. However, none of the studies explicitly accounted for ancestry, which has been shown to influence transcriptomic variation (58). Future work must model ancestry in transcriptomic analyses, especially in admixed populations.

Epigenetic studies (mainly DNA methylation) provide valuable insights into how environmental exposures may shape suicide risk at the molecular level (59). These studies are particularly relevant given the socioeconomic and environmental diversity across and within Latin American countries (60). Altered methylation patterns were often observed in genes involved in stress response, neurodevelopment, and brain function. Some studies also found correlations between DNA methylation and gene expression, suggesting cross-level interactions that may contribute to suicide risk. Integrative analyses combining PGS and DNA methylation data suggest that genetic liability may influence epigenetic profiles (48), reinforcing the importance of multi-omics approaches. Future studies should incorporate social determinants of health, including trauma, discrimination, and migration-related stress, which may disproportionately affect Latin American and Hispanic/Latinx populations (61).

Despite recent advances, major gaps remain. Many studies, particularly candidate gene investigations, had small sample sizes, limiting both statistical power and reproducibility. A persistent limitation across studies was the lack of population stratification control and ancestry-specific analyses. Latin America is highly admixed, and failure to model ancestry can result in spurious associations (12). Including Latin American populations in genomic research is necessary, but not sufficient, unless appropriate analytical methods tailored to admixed populations are used. Tools such as Tractor (62) and local ancestry-adjusted allelic association analysis (63) offer promising frameworks that should be adopted in future studies. A striking observation from this review is that no studies to date have examined NSSI despite its high burden in adolescents (64), a demographic group that comprises a large portion of the Latin American population. This omission is especially concerning and represents an important direction for future research.

Network analysis showed that the field is dominated by one collaborative group (mainly U.S./Mexico), whereas many regional researchers remain disconnected. The interaction among these nodes defines the transnational architecture of the system, where international collaboration occurs in a mediated and asymmetrical fashion; connections between countries exist, but they are channeled through specific intermediaries rather than through a dense web of collaborations. Altogether, the observed structure suggests a hierarchical coauthorship system in which many peripheral

participants depend on a few coordinating nuclei. This hierarchical pattern favors international visibility but may limit horizontal knowledge transfer, reduce exposure of smaller groups to international consortia, and perpetuate inequities in scientific visibility and resource access. Limited collaboration likely reflects disparities in research infrastructure, funding, language barriers, and access to international consortia. Increasing regional and international collaboration, sharing data, and leveraging existing samples for secondary analyses could improve statistical power and foster innovation. Strengthening cross-institutional links, particularly among less-connected centers, will be essential to promote the sustainable growth of suicide genomics in Latin America. Encouragingly, molecular research in the region is growing. Nonetheless, expanding visibility and supporting local capacity-building and equitable inclusion of Latin American researchers in global efforts are urgently needed (65). As part of these efforts, Latin American researchers have established the Suicide Working Group within the Latin American Genomics Consortium, aiming to advance research in this field and foster international collaboration (4).

This study has some limitations that should be noted. First, despite extensive searches, relevant studies, particularly those published in nonindexed journals, may have been missed. Second, several studies identified in this review used overlapping samples, reflecting the limited availability of postmortem tissue and biobanks in the region, which constrains opportunities for replication and cross-validation of findings. Expanding sample collection through collaboration with coroners, brain banks, and clinical networks will diversify data sources, increase representativeness, and enable replication of genomic findings in the region. Additionally, quantitative meta-analysis was limited by such overlapping samples, as well as heterogeneous designs, the absence of overlapping variants, inconsistent phenotype definitions, and incomplete reporting. Furthermore, the heterogeneity in suicidality phenotypes also constrained integration. Standardized operationalization of suicidality phenotypes in genomic and epigenomic research is needed to enable more precise meta-analytic and mechanistic analyses (66).

To move the field forward, future studies should prioritize the recruitment of larger, locally representative, and ancestrally diverse samples. They should also incorporate ancestry-aware statistical models and combine genomic, epigenetic, and transcriptomic data with environmental and clinical information. Such efforts will help clarify the biological pathways involved in suicide risk and support the development of more inclusive, effective prevention strategies. Studying admixed populations such as those in Latin America is not only essential for addressing local health disparities, but it is also an opportunity to enrich global psychiatric research (67). The region's genetic and environmental complexity offers a powerful lens through which to identify shared and unique pathways to suicide risk. Therefore, findings from Latin American studies can inform models of suicidality that are more robust and globally relevant.

CONCLUSIONS

The reviewed literature highlights the multifactorial biological nature of suicidal behaviors and points to potential

ancestry-specific genetic risk factors. Addressing current limitations requires investment in the inclusion of Latin American populations in psychiatric genomics. This includes prioritizing local recruitment, using appropriate analytical tools, multi-omics integration, and contextual data. Building regional research capacity and fostering equitable international collaborations will be key to producing findings that are scientifically valid, socially meaningful, and globally relevant.

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If you are contemplating suicide, please reach out for help; resources are available at <https://www.iasp.info/suicidalthoughts/>.

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